

UNIT #3

Applying Properties of Derivatives

Curve Sketching

Ex) If $f(x) = x^4 + 12x^3 - 2x^2 - 420x$, find the intervals of increase or decrease. Hint:
 $f'(3) = 0$ (use long division)

Ex) If $f(x) = -3x^{\frac{5}{3}} + 15x^{\frac{2}{3}}$ find the intervals of increase or decrease.

Algebraic Procedure to finding Local Maximum and Minimum Points

1. Find $f'(x)$ and factor it
2. Determine all of the critical numbers by finding all numbers that make $f'(x) = 0$ or $f'(x)$ undefined.
3. Order your critical numbers from most negative to most positive and use them to create an interval chart.
4. Complete your interval chart and use it to determine if $f'(x)$ changes its sign at c . Note if it's a local min or a local max by viewing the type of sign change of $f'(x)$.
5. If a local min or local max exists at c then determine $f(c)$ to find the y coordinate of your local min or local max point.

If you are asked to determine Absolute Maximum and Minimum points in an interval $x \in [a, b]$ then

6. Evaluate $f(a)$ and $f(b)$ and compare the end points of the interval with the local max and local min points you found in steps 1-5.

Ex) Given $f(x) = 2x^3 - 3x^2 - 12x + 5$ determine

- a) The critical numbers of $f'(x)$
- b) The Local Max and Local Min points
- c) The Absolute Max and Absolute Min points in the interval $x \in [-2, 4]$
- d) Make a rough sketch of $f(x)$

Ex) Given $f(x) = 4x^3 + 3x^2 - 6x + 1$ determine

- a) The Local Max and Local Min points
- b) Make a rough sketch of $f(x)$

Ex) Given $f(x) = (1 - x^2)^2 - 2$ determine

- a) The Local Max and Local Min points
- b) Make a rough sketch of $f(x)$

Ex) For the cubic function $f(x) = ax^3 + bx^2 + cx + d$, find the values of a , b , c , and d such that there is a local maximum at $(-2, 13)$ and a local minimum at $(0, 5)$. [$a = 2, b = 6, c = 0, d = 5$]

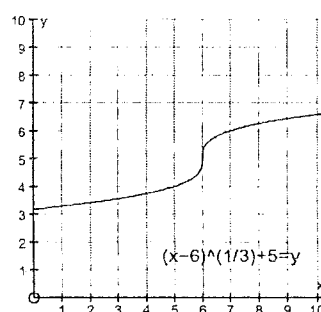
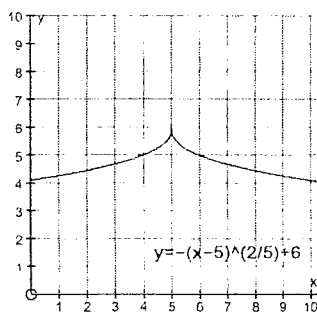
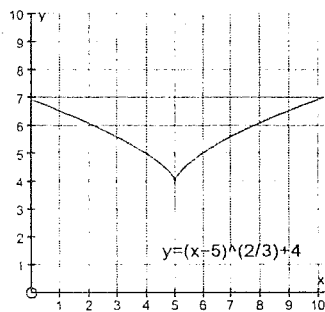
(i) Given $f(x) = \frac{x^2 - x + 1}{x^2 + 1}$ determine the Local Max and Local Min points $\left[-1, \frac{3}{2}\right], \left(1, \frac{1}{2}\right)$

Ex) If $f(x) = x^4 - 8x^3$

- Classify all local extrema using the Second Derivative Test. If the Second Derivative Test fails, use the First Derivative Test.
- Determine the Points of Inflection by creating an interval chart for $f''(x)$

A word on Cusps

A cusp is a pointy bend on a graph where no tangent line exists. A cusp is a good example of the situation where a function can have a maximum or minimum at a point where the derivative of the function does not exist.



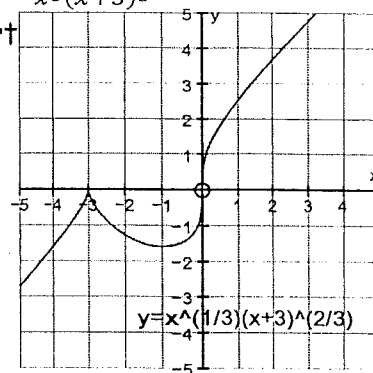
To spot a cusp

- Determine where the derivative does not exist and test the concavity on either side of that critical number.
- If the concavity remains the same, you have found a cusp (first 2 pictures). If the concavity changes, you have found a P.O.I with a vertical tangent (3rd picture).

Ex) Verify that there is a cusp on the graph of the function $f(x) = x^{2/3}$ when $x = 0$.

Ex) For the function $f(x) = x^{\frac{1}{3}}(x+3)^{\frac{2}{3}}$, $f'(x) = \frac{x+1}{x^{\frac{2}{3}}(x+3)^{\frac{1}{3}}}$, and $f''(x) = -\frac{2}{x^{\frac{5}{3}}(x+3)^{\frac{4}{3}}}$

- Find the intervals of increase and decrease using an interval chart
- Find the local maximum and minimum points
- Find the intervals of concavity using an interval chart
- Find the points of inflection
- Label the points you found in b) and d) on the graph to the right

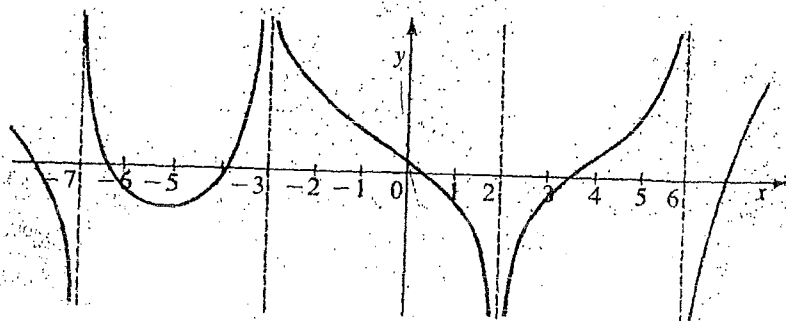


Ex) For the curve $f(x) = 2 - 12x + 9x^2 - 2x^3$.

- Find the local maximum and minimum points using an interval chart $[(1, -3) \text{ and } (2, -2)]$
- Find any points of inflection using an interval chart $[\frac{3}{2}, -2.5]$

Vertical Asymptotes.

1. The graph of f is given.



(a) State the equations of the vertical asymptotes.

(b) State the following.

(i) $\lim_{x \rightarrow -7^-} f(x)$

(ii) $\lim_{x \rightarrow -7^+} f(x)$

(iii) $\lim_{x \rightarrow -3} f(x)$

(iv) $\lim_{x \rightarrow 2} f(x)$

(v) $\lim_{x \rightarrow 6^-} f(x)$

(vi) $\lim_{x \rightarrow 6^+} f(x)$

2. Find the vertical asymptotes and sketch the graph near the asymptotes.

(a) $y = \frac{2}{x+1}$

(b) $y = \frac{3}{(x-6)^2}$

(c) $y = \frac{x}{(x+2)^2}$

(d) $y = \frac{1}{x^2-1}$

(e) $y = \frac{x}{x^2-1}$

(f) $y = \frac{6x^3}{x^2+4x+3}$

(g) $y = \frac{1}{x^2(x+1)}$

(h) $y = \frac{1}{x^4-4x^2}$

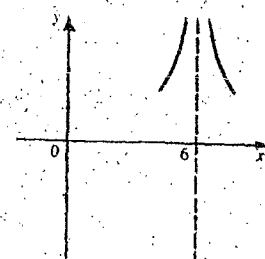
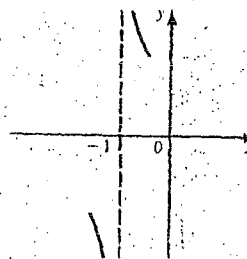
Answers:

1. (a) $x = -7, x = -3, x = 2, x = 6$

(b) (i) $-\infty$ (ii) ∞ (iii) ∞ (iv) $-\infty$

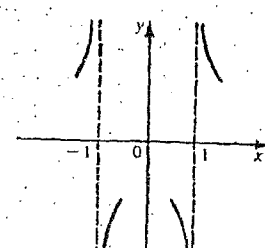
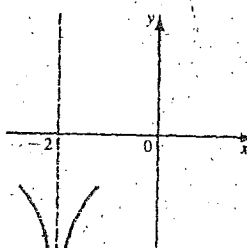
(v) ∞ (vi) $-\infty$

3. (a) $x = -1$ (b) $x = 6$



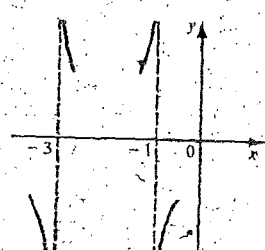
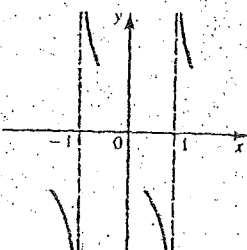
(c) $x = -2$

(d) $x = -1, x = 1$



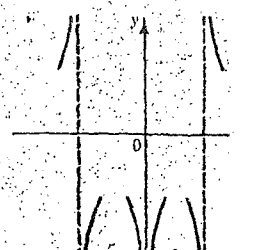
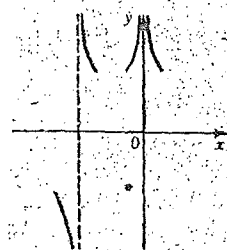
(e) $x = -1, x = 1$

(f) $x = -3, x = -1$



(g) $x = -1, x = 0$

(h) $x = -2, x = 0, x = 2$



Sketching Practice

Sketch each of the following using Intercepts, Asymptotes, Critical Points, Points of Inflection and Sign Charts.

a) $y = \frac{1}{x}$

b) $y = \frac{1}{x+2}$

c) $y = x^2(x-1)$

d) $y = x(2x-1)(x+1)$

e) $y = \frac{x^3}{1-x}$

f) $y = \frac{1-x}{1+x^2}$

g) $y = \frac{x^2-1}{x^2-4}$

h) $y = \frac{x}{(x-1)(x-4)}$

i) $y = \frac{x}{x^2+1}$

j) $y = \frac{1}{(x-5)(x-1)}$

k) $y = \frac{x^2}{x^2+4}$

l) $y = \frac{x}{x^2-1}$

m) $y = x - \frac{1}{x}$

n) $y = \frac{x^2}{x+1}$

o) $y = \frac{x}{x+1}$

p) $y = \frac{2(x-4)}{x^2}$

q) $y = \frac{x^2}{x+2}$

r) $y = \frac{x^2+2x+3}{x^2+3x+2}$

s) $y = \frac{x^2-7x+16}{x-4}$

t) $y = x^2 + \frac{2}{x} - 2$

u) $y = \frac{x^3}{3x+6}$

v) $y = \frac{x-1}{x(x+1)}$

Curve Sketching – Putting it all together - 1

Follow the algorithm for curve sketching to sketch $f(x) = \frac{2x^2}{x^2-1}$ given that $f'(x) = \frac{-4x}{(x-1)^2(x+1)^2}$ and $f''(x) = \frac{4(3x^2+1)}{(x-1)^3(x+1)^3}$

Domain: _____

Intercepts: x-int(s): _____

y-int _____

Asymptote(s):

Horizontal _____

Vertical _____

First Derivative Test

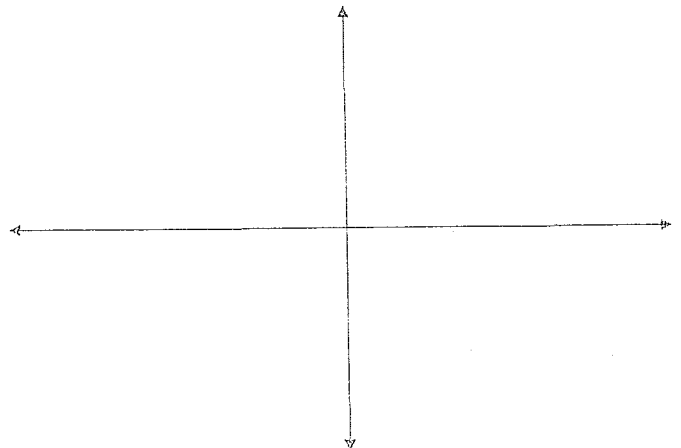
Critical number(s): _____

Local Extrema: Maxima: _____

Minima: _____

Test for Concavity Point(s) of Inflection: _____

Sketch of Curve



Curve Sketching – Putting it all together - 2

Follow the algorithm for curve sketching to sketch $f(x) = \frac{x^2-16}{x^2-4}$ given that $f'(x) = \frac{24x}{(x-2)^2(x+2)^2}$ and

$$f''(x) = \frac{-24(3x^2+4)}{(x-2)^3(x+2)^3}$$

Domain: _____

Intercepts: x-int(s): _____

y-int _____

Asymptote(s):

Horizontal _____

Vertical _____

First Derivative Test

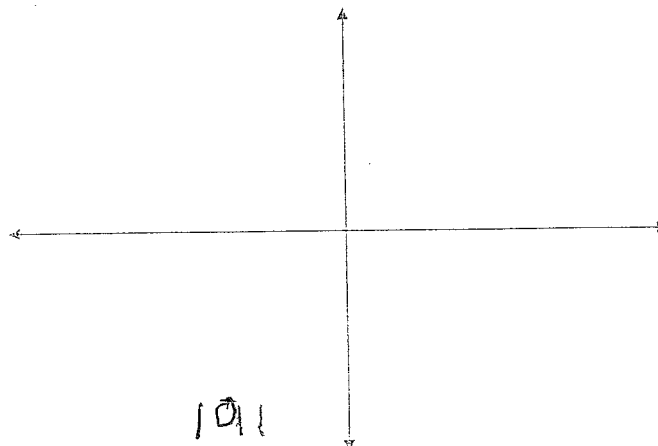
Critical number(s): _____

Local Extrema: Maxima: _____

Minima: _____

Test for Concavity Point(s) of Inflection: _____

Sketch of Curve



Curve Sketching – Putting it all together - 3

Follow the algorithm for curve sketching to sketch $f(x) = x^{\frac{2}{3}}(x - 5)$ given that $f'(x) = \frac{5(x-2)}{3\sqrt[3]{x}}$ and $f''(x) = \frac{10(x+1)}{9x^{\frac{4}{3}}}$

Domain: _____

Intercepts: x-int(s): _____

y-int _____

Asymptote(s):

Horizontal

Vertical

First Derivative Test

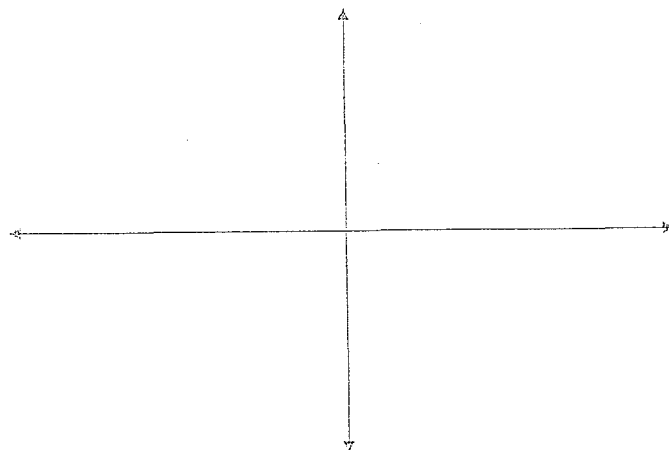
Critical number(s): _____

Local Extrema: Maxima: _____

Minima: _____

Test for Concavity Point(s) of Inflection: _____

Sketch of Curve



Curve Sketching - Putting it all together - 4

Follow the algorithm for curve sketching to sketch $f(x) = (x-1)^3(x+1)^3$ given that $f'(x) = 6x(x-1)^2(x+1)^2$ and $f''(x) = 6(x-1)(x+1)(5x^2-1)$

Domain: _____ Intercepts: x-int(s): _____ y-int: _____

Asymptote(s):

Horizontal

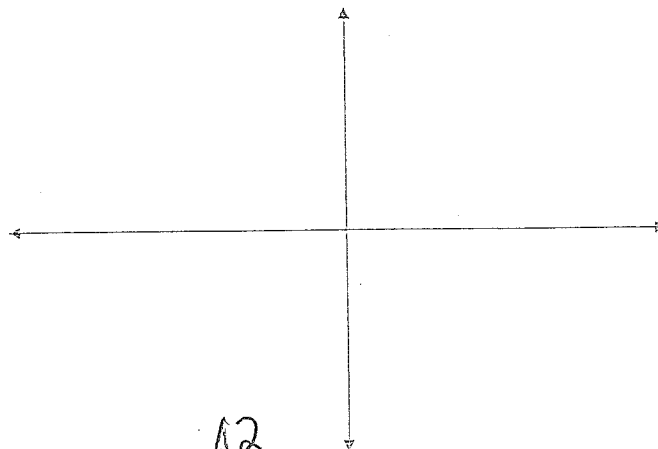
Vertical

First Derivative Test

Critical number(s): _____ Local Extrema: Maxima: _____ Minima: _____

Test for Concavity Point(s) of Inflection: _____

Sketch of Curve



Curve Sketching – Putting it all together - 5

Follow the algorithm for curve sketching to sketch $f(x) = x\sqrt{8-x^2}$ given that

$$f'(x) = \frac{-2(x-2)(x+2)}{\sqrt{8-x^2}} \text{ and } f''(x) = \frac{2x(x+2\sqrt{3})(x-2\sqrt{3})}{(x^2+8)^{\frac{3}{2}}}$$

Domain: _____

Intercepts: x-int(s): _____

y-int: _____

Asymptote(s):

Horizontal

Vertical

First Derivative Test

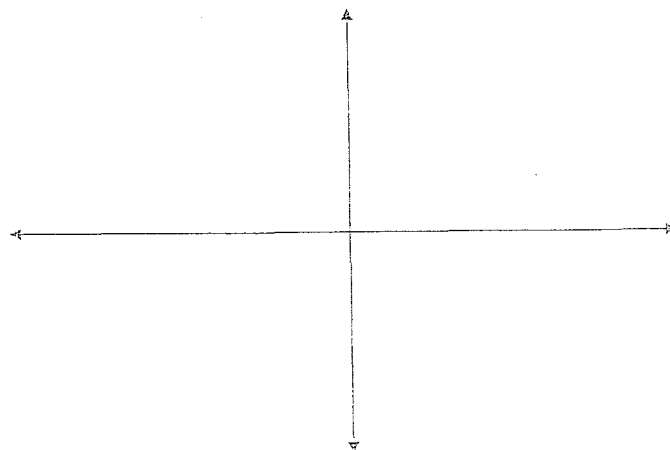
Critical number(s): _____

Local Extrema: Maxima: _____

Minima: _____

Test for Concavity Point(s) of Inflection: _____

Sketch of Curve



Curve Sketching – Putting it all together - 6

Follow the algorithm for curve sketching to sketch $f(x) = (x - 3)^2 \cdot \sqrt[3]{x + 5} + 5$ given that

$$f'(x) = \frac{(x-3)(7x+27)}{3(x+5)^{\frac{2}{3}}} \text{ and } f''(x) = \frac{4(7x^2+54x+63)}{9(x+5)^{\frac{5}{3}}}$$

Domain: _____

Intercepts: x-int(s): _____

y-int _____

Asymptote(s):

Horizontal _____

Vertical _____

First Derivative Test

Critical number(s): _____

Local Extrema: Maxima: _____

Minima: _____

Test for Concavity Point(s) of Inflection: _____

Sketch of Curve

